



KPROJECTXX PRESENTS

HACKFUSION

SOFTWARE DOMAIN PROBLEM STATEMENTS

WEB & MOBILE APPLICATIONS

- **Integrated Multi-Tier Enterprise Platform:** A unified ecosystem combining product e-commerce, custom prototyping workflows, and training modules with multi-tier role-based access. Designed to streamline complex operations across diverse organizational structures seamlessly.
- **Cross-Platform Canteen Management Ecosystem:** A synchronized platform bridging reception billing and stall fulfillment with real-time inventory tracking to prevent out-of-stock orders. Ensures fast processing times and continuous supply chain visibility during peak hours.

AI/ML & COMPUTER VISION

- **AI-Powered Resume ATS Optimization Engine:** An intelligent parsing tool that evaluates resumes against Job Descriptions, calculates selection probability, and provides a skill-gap roadmap. Helps job seekers align their profiles precisely with market demands.
- **AI-Powered Cattle Health Monitoring Ecosystem:** A non-invasive computer vision system using CCTV feeds to track cattle behavior, predict estrus cycles, and detect early signs of diseases. Enables proactive livestock management without requiring intrusive wearable devices.
- **Privacy-First Offline Document Parsing Engine:** A secure, offline object detection and OCR pipeline to extract specific fields from sensitive KYC documents without cloud dependency. Protects user data privacy while maintaining high field-extraction accuracy locally.

AUTOMATION & DEVELOPER TOOLS

- **Autonomous SLAM-Driven Rover Navigation Pipeline:** An edge-computed SLAM system that maps unknown environments and dynamically navigates to user-defined coordinates or patterns without manual teleoperation. Offers reliable positioning and obstacle avoidance in offline settings.
- **AI-Driven Kinematic Calculator for Unmanned Rovers:** A developer tool that calculates exact torque and RPM requirements based on rover dimensions and terrain, using AI to recommend optimal motor configurations. Accelerates hardware selection and hardware prototyping cycles.